

Fizik 1 Tekrar

3 temel büyüklük vardır.

- 1- Kütle (kg) 3- Zaman (s)
2- Uzunluk (m)

Boyut Analizi

<u>Büyüklik</u>	<u>Boyut</u>	<u>Birim</u>
Kütle	Mass	kg
Uzunluk	Length	metre
Zaman	Time	saniye
Alan	L^2	m^2
Hacim	L^3	m^3
Hız	L/T	m/s
İvme	L/T^2	m/s^2
Kuvvet	$M \cdot L/T^2$	$kg \cdot m/s^2$

* Bir eşitliğin her iki tarafının boyutları aynı olmalıdır.

* Aynı boyuttaki nicelikler toplanıp çıkarılabilir.

Ör/

$$F = G \cdot \frac{m_1 \cdot m_2}{r^2}$$

"G" boyut analizini yapınız.

$$\frac{M \cdot L}{T^2} = "G" \cdot \frac{M^2}{L^2}$$

$$\frac{L^3}{M \cdot T^2} = "G"$$

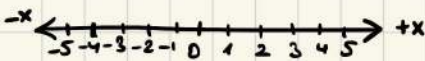
Birim Çevirme

$$1. \frac{1 \text{ km}}{1000 \text{ m}} \cdot \frac{3600 \text{ s}}{1 \text{ h}}$$

$$1 \text{ m/s} = ? \text{ km/h}$$

$$1 \frac{\text{m}}{\text{s}} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) \cdot \left(\frac{1 \text{ km}}{1000 \text{ m}} \right) = 3,6 \text{ km/h}$$

Bir Boyutlu Hareket



1) Yer değişimi: $\Delta x = x_s - x_i$ (metre)

x_i = ilk konum	t_i = ilk zaman
x_s = son konum	t_s = son zaman

2) Ortalama Hız: $v_{\text{ort}} \equiv \frac{\Delta x}{\Delta t}$ (m/s)

3) Ani Hız: $\lim_{\Delta t \rightarrow 0} v_{\text{ort}} \equiv v$

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

4) Ortalama ivme:

$$a_{\text{ort}} \equiv \frac{\Delta v}{\Delta t} = \frac{v_s - v_i}{t_s - t_i} \quad (\text{m/s}^2)$$

5) Ani ivme

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv(t)}{dt}$$

Türev Operatörü

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta}{\Delta t} \equiv \frac{d}{dt}$$

$$\text{Ör/ } -4t + 2t^2 = x$$

a) 1sn ile 3sn arasında yer değiştirme

$$\begin{array}{l} 4t - 4 = 12 \\ 4 = 9 \end{array} \quad \left/ \quad \begin{array}{l} (t=1) \quad x_i = -2\text{m} \\ (t=3) \quad x_s = 6\text{m} \end{array} \right) 8\text{m}$$

b) Ort hız:

$$8/2 = 4\text{ m/s}$$

$$\frac{8-0}{2} = 4\text{ m/s}$$

c) $t=2.5\text{sn'de}$ cismin hızı

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{x_s - x_i}{\Delta t}$$

$$t_s = t + \Delta t$$

$$x_s = -4(t + \Delta t) + 2(t + \Delta t)^2$$

$$v = \frac{dx}{dt} \bigg|_{t=2.5} \rightarrow \begin{array}{l} -4 + 4t \\ 6\text{m} \end{array}$$

$$\lim_{\Delta t \rightarrow 0} \frac{-\cancel{4t} - 4\Delta t + 2\cancel{t^2} + 4t\Delta t + 2\Delta t^2 + \cancel{4t} - 2\cancel{t^2}}{\Delta t}$$

$$\frac{\cancel{\Delta t}(-4 + 4t + 2\Delta t)}{\cancel{\Delta t}} = -4 + 4t \quad (t=2.5) = 6$$

Sabit İvmeli Hareket ($a = sbt$)

- Ortalama ivme, ani ivmeye eşittir.

Düzgün ivmede geçerlidir.

$$v(t) = v_i + at$$

+ anında v hızı
ilk hızı + ivme $\times$ zaman

$$v_{ort} = \frac{x(t) - x_i}{t} = \frac{v_i + v(t)}{2}$$

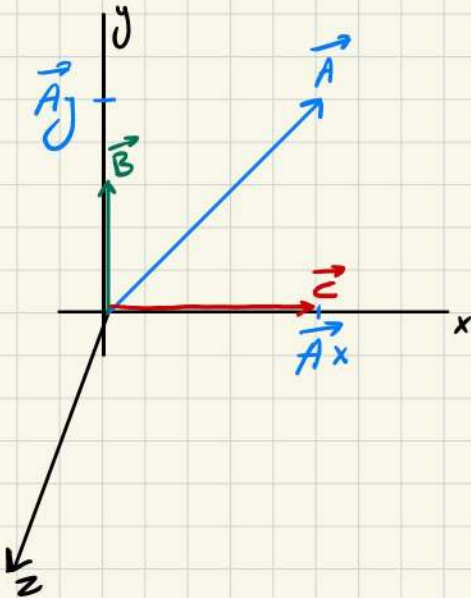
$$x(t) = x_i + v_i t + \frac{1}{2} at^2$$

$$x_s = x_i + v_i t + \frac{at^2}{2}$$

$$v^2 = v_i^2 + 2a(x - x_i)$$

Zamansız hız formülü

Vektörler



Birim Vektör: $|\hat{i}| = 1 \text{ br}$

$|\hat{j}| = 1 \text{ br}$

$|\hat{k}| = 1 \text{ br}$

$$\vec{B} = 3\hat{i}$$

$$\vec{C} = 4\hat{j}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

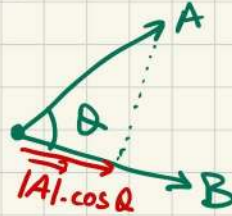
$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

↓
 A 'nın uzunluğu

Vektörlere Çarpım

1) Skaler Çarpım (iç çarpım)

$$\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos \theta$$



$$\hat{i} \cdot \hat{i} = |\hat{i}| \cdot |\hat{i}| \cdot \cos 0^\circ = 1$$

$$\hat{j} \cdot \hat{j} = |\hat{j}| \cdot |\hat{j}| \cdot \cos 0^\circ = 1$$

$$\hat{h} \cdot \hat{h} = 1$$

$$\hat{i} \cdot \hat{j} = |\hat{i}| \cdot |\hat{j}| \cdot \cos 90^\circ = 0$$

$$\hat{i} \cdot \hat{h} = 0$$

$$\hat{j} \cdot \hat{h} = 0$$

$$\vec{A} \cdot \vec{B} = A_x B_x \hat{i} \cdot \hat{i} + A_x B_y \hat{i} \cdot \hat{j} + A_x B_z \hat{i} \cdot \hat{k} + A_y B_x \dots$$

↓

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Ör/

$$\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$$
$$\vec{B} = 3\hat{i} - 4\hat{j} + 5\hat{k}$$

$$a) 6 - 12 - 5 = -11$$

$$6 - 12 - 5 = -11$$

$$-11 = \sqrt{14} \cdot \sqrt{50} \cos \theta$$

$$\cos^{-1}\left(\frac{-11}{\sqrt{14} \cdot \sqrt{50}}\right) = \theta$$

a) A ve B skaler çarpım

b) A, B arası açı kaç derecedir?

$$b) \sqrt{3^2 + (-4)^2 + 5^2} = 7.07$$

$$\sqrt{2^2 + 3^2 + (-1)^2} = 3.74$$

$$-11 = (7.07)(3.74) \cdot \cos \theta$$

$$\cos \theta = -0.38$$

$$\theta = 112.3^\circ$$

2) Vektörel Çarpım

$$\vec{A} \times \vec{B}$$

Vektörel Çarpımın Özellikleri

1) İki vektör çarpımı da bir vektördür.

$$2) |\vec{C}| = |\vec{A} \times \vec{B}| = |\vec{A}| \cdot |\vec{B}| \cdot \sin \theta$$

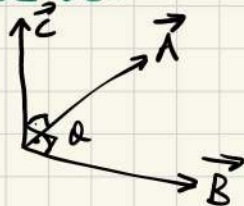
3) İki vektör çarpımı diğer vektörlere diktir.

Yani aynı düzlemde değildir.

$$|\hat{i} \times \hat{i}| = |\hat{i}| \cdot |\hat{i}| \cdot \sin 0^\circ = 0$$

$$|\hat{j} \times \hat{j}| = |\hat{j}| \cdot |\hat{j}| \cdot \sin 0^\circ = 0$$

$$4) \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$



$$\hat{i} \times \hat{j} = \hat{z}$$

$$\underline{\hat{i} \cdot \hat{j} = \hat{k}}$$

$$\vec{z} = |\hat{i}| \cdot |\hat{j}| \cdot \sin 90^\circ = 1$$

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

$$\vec{A} = 2\hat{i} + 3\hat{j}$$

$$\vec{B} = \hat{i} - 5\hat{j}$$

$$\vec{A} \times \vec{B} = (2\hat{i} + 3\hat{j}) \times (\hat{i} - 5\hat{j})$$

Diagram showing the expansion of the cross product using the distributive property:

- $2\hat{i} \times \hat{i} = 0$
- $2\hat{i} \times (-5\hat{j}) = -10\hat{k}$
- $3\hat{j} \times \hat{i} = -3\hat{k}$
- $3\hat{j} \times (-5\hat{j}) = 0$

$$\vec{A} \times \vec{B} = -13\hat{k}$$

$$|\vec{A} \times \vec{B}| = 13$$

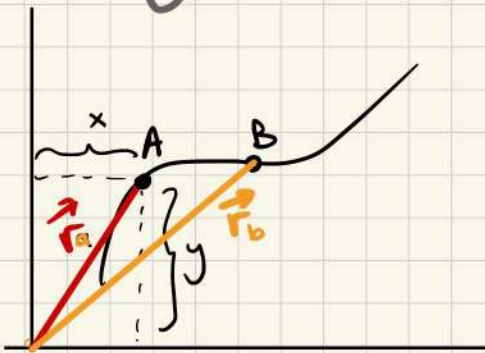
$$(\vec{A} \times \vec{B}) \cdot \vec{A} = ?$$

$$\underline{\underline{0}}$$

$$\vec{z} \perp \vec{A}$$

$$\vec{z} \perp \vec{B}$$

İki Boyutlu Hareket



$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$
$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

1) Yer değiştirme

$$\Delta \vec{r} = \vec{r}_B - \vec{r}_A = (x_B \hat{i} + y_B \hat{j}) - (x_A \hat{i} + y_A \hat{j})$$

2) Ortalama Hız

$$\vec{v}_{\text{ort}} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta x}{\Delta t} \hat{i} + \frac{\Delta y}{\Delta t} \hat{j} + \frac{\Delta z}{\Delta t} \hat{k}$$

$$\vec{v}_{\text{ort}} = v_{\text{ort}x} \hat{i} + v_{\text{ort}y} \hat{j} + v_{\text{ort}z} \hat{k}$$

3) Anlık Hız

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

$$\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$$

4) Ortalama ivme

$$\vec{a}_{\text{ort}} = \frac{\Delta \vec{v}}{\Delta t} = a_{\text{ort}x} \hat{i} + a_{\text{ort}y} \hat{j}$$

5) Ani ivme

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2} \quad \vec{a} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j}$$

Sabit ivmeli iki Boyutlu Hareket

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$v_x = v_{0x} + a_x t$$

$$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a} t^2$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$v_y = v_{0y} + a_y t$$

a) $x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$
 $x = 20t + \frac{1}{2}(4)t^2$
 $x = 20t + 2t^2$
 $4t^2 + 20t$
 $(2t+5)^2 = (10)^2$

b) $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$
 $y = -15t + \frac{1}{2}(4)t^2$
 $y = -15t + 2t^2$
 $2t^2 - 15t$
 $t(2t-15) = 0$
 $t = 0$ or $t = 7.5$

c) $v_x = v_{0x} + a_x t$
 $v_x = 20 + 4t$
 $v_x = 20 + 4(7.5)$
 $v_x = 50$
 $v_y = v_{0y} + a_y t$
 $v_y = -15 + 4t$
 $v_y = -15 + 4(7.5)$
 $v_y = 15$

$$a_x = 4 \text{ m/s}^2$$

$t=0$ anında orijinde başlayan

$$a_y = 0$$

$$v_{0x} = 20 \text{ m/s}$$

$$v_{0y} = -15 \text{ m/s}$$

a) Herhangi bir andaki hız vek. yazınız.

b) $(t=5)$ için hız, sürat x eksenini ile açısı

c) Konum vek?

$$c) \vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a} t^2$$

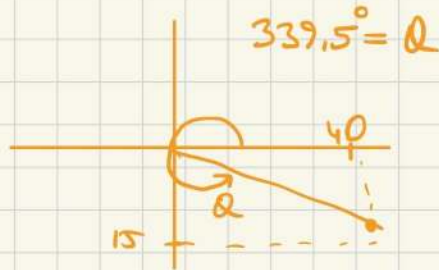
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$$20\hat{i} - 15\hat{j}$$

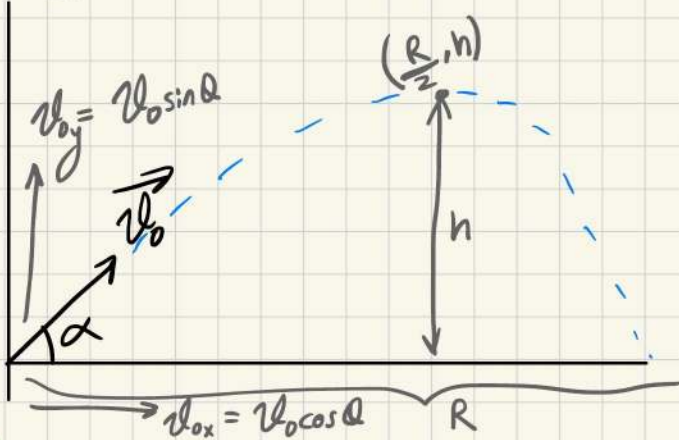
$$\vec{a} = 4\hat{i}$$

$$\vec{v} = \vec{v}_0 + \vec{a}t$$

$$\vec{v} = (20\hat{i} - 15\hat{j}) + (4\hat{i}t)$$



Yatay Atış Hareketi



$$a = -g = -9.8 \frac{m}{s^2}$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$x = v_0 \cos \alpha \cdot v_0 \sin \alpha \cdot 2$$

$$x = \frac{v_0^2 \sin(2\alpha)}{g}$$

Maksimum uzaklık

1) Cismin ivmesi "g" ve sbt

2) Hava Direnci ihmal

$$v_y = v_{0y} + a_y t$$



$$v_y = v_0 \sin \alpha - gt$$

+ anında
hızın y
eksenindeki
değeri

tepe noktasında $v_y = 0$

$$0 = v_0 \sin \alpha - gt_c$$

$$t_c = \frac{v_0 \sin \alpha}{g} = t_i$$

çıkış = iniş süresi

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$t = t_c$ olursa $y = h$

$$h = v_0 \sin \alpha \cdot \frac{v_0 \sin \alpha}{g} - \frac{1}{2}g \frac{v_0^2 \sin^2 \alpha}{g^2}$$



$$h = \frac{v_0^2 \sin^2 \alpha}{2g}$$

Maksimum yükseklik

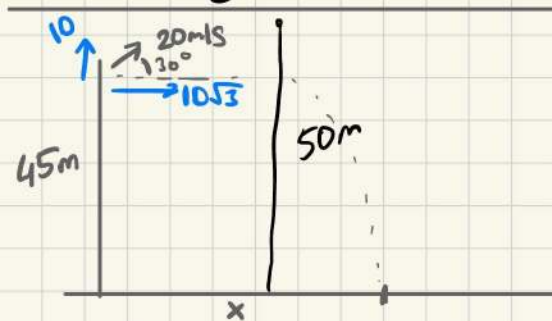
Bir taş yerle 20° atılarak ($V_0 = 11 \text{ m/s}$)

a) Yatayda ne kadar uzam alır?

$$x = \frac{v_0^2 \sin 40^\circ}{g} = \frac{121 \cdot \sin 40^\circ}{9.8} = 7.93 \text{ metre}$$

b) Tasın maks yükseklik

$$h = \frac{v_0^2 \cdot \sin^2 \alpha}{2g} = \frac{121 \cdot \sin^2 20^\circ}{9,8} = 0,72 \text{ m}$$



a) tad ne zaman yere qapar.

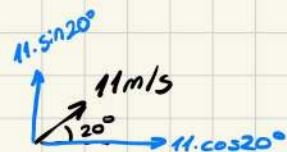
$$t_q = 150$$

$$h_{max} = 45 + 10 - 5 = 50 \quad | \quad 3.16 + 1 = \underline{\underline{4.16}}$$

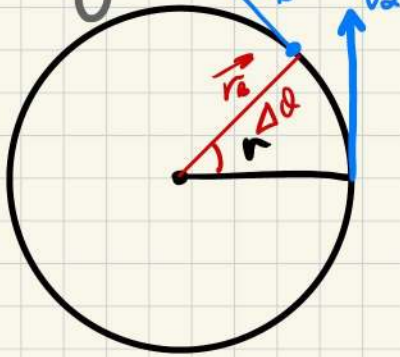
b) tasin yere carpan hizi nedir?

31.35 m/s

$$10.1 - \frac{1 \cdot 10.1}{2} = \underline{\underline{5m}}$$

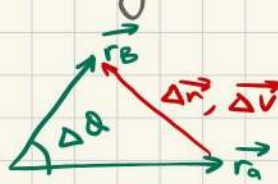


Düzgün Dairesel Hareket



$$\vec{a} = \frac{d\vec{v}}{dt}$$

merkezcil ivme: dairenin merkezine doğru olan ivme



$$|\vec{v}_A| = |\vec{v}_B| = v$$

$$|\vec{r}_A| = |\vec{r}_B| = r$$

$$\frac{|\Delta \vec{r}|}{r} = \frac{|\Delta \vec{v}|}{v}$$

$$|\vec{a}_{ort}| = \frac{|\Delta \vec{v}|}{\Delta t}$$

$$v = \lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{r}|}{\Delta t} = \frac{|\dot{\vec{r}}|}{dt} \left\{ \text{anlık hız} \right.$$

$$|\vec{a}_{anlık}| = \frac{v^2}{r} \left\{ \text{merkezcil ivme} \right.$$

$$\vec{a}_r + \vec{a}_\theta = \vec{a}_t$$

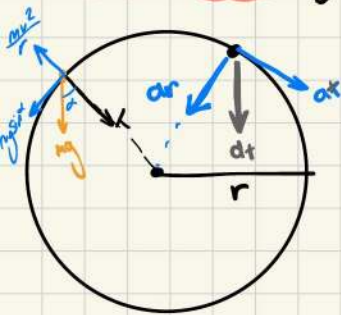
Düzgün olmayan dairesel hareket

1- Hızın büyüklüğünün değişmesinden
ivme

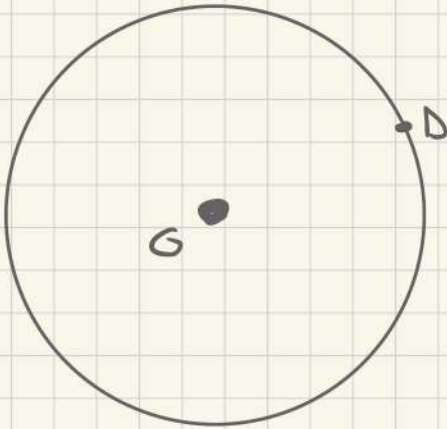
$$a_1 = \frac{dv}{dt}$$

2- Hızın yönünden dolayı ivme

$$a_2 = \frac{v^2}{r}$$



Dünyanın, güneş etrafında dönmesi sonucu merkezî ivmesi yaklaşık nedir?



$$a = \frac{v^2}{r}$$

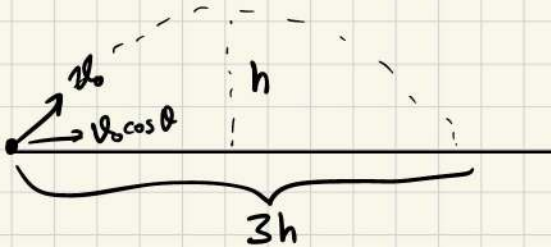
$$\underbrace{2\pi r}_{\text{yol}} = v \cdot t$$

$$\frac{2\pi r}{t} = v$$

$$a = \frac{4\pi^2 r^2}{t^2 \cdot r}$$

$$a = 5.93 \cdot 10^{-3} \text{ m/s}^2$$

Bir top yenden belli bir açıyla fırlatıldığında menzil uzaklığı yüksekliğ'in 3 katı oluyor. Açı=?



$$\frac{v_0^2 \cdot \sin 2\theta}{g} = 3h$$

$$\frac{3 v_0^2 \cdot \sin^2 \theta}{2g} = 3h$$

$$\frac{3 \cancel{v_0^2} \cdot \sin^2 \theta}{\cancel{2g}} = \frac{\cancel{v_0^2} \cdot \sin 2\theta}{g}$$

$$3 \sin^2 \theta = 2 \sin 2\theta$$

$$3s^2 = 2 - 4s^2$$

$$7s^2 = 2$$

$$s = \pm \sqrt{\frac{2}{7}}$$

Orijinde

$$\vec{a} = 3\hat{j} \text{ m/s}^2$$

$$\vec{v}_0 = 5\hat{i} \text{ m/s}$$

ise

a) herhangi anda konum vek

$$x = v_0 \cdot t + \frac{1}{2} a t^2$$

$$r = 5\hat{i}t + \frac{3\hat{j}}{2}t^2$$

b) hız vek

$$v = 5\hat{i} + 3\hat{j}t = \frac{dr}{dt}$$

c) $t=2$ için koordinat, sürat

$$(10\hat{i}, 6\hat{j}) \quad / \quad 5\hat{i} + 6\hat{j} = \vec{v}$$

$$\sqrt{25+36} = |\vec{v}|$$

$$\text{m/s} \cdot 7.81 = |\vec{v}|$$

Hareket Yasaları

Kuvvet: $\vec{F}$

1- Eylemsizlik

1) Temas Kuvvetleri

2) Alan Kuvvetleri

$$\sum \vec{F} = 0$$
$$\vec{a} = 0$$

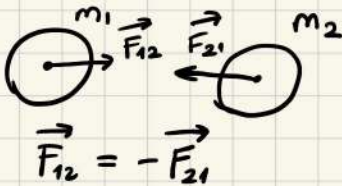
Kütle: Cismin hızının değişimine karşı gösterdiği direnç

2- $F = m \cdot a$ (kg/m.s^2)

$$m_1 \rightarrow a_1 \quad \frac{m_1}{m_2} = \frac{a_2}{a_1}$$

$$m_2 \rightarrow a_2$$

3- Etki/Tepki Prensibi



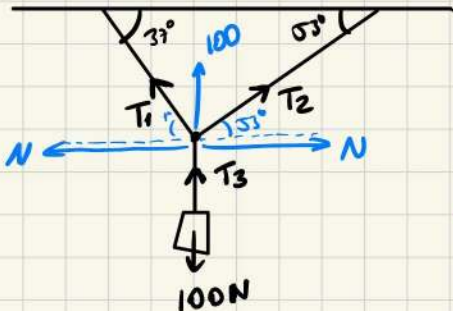
4- Kütle Çekim Kuvveti

$$\vec{F}_g \quad \vec{a} = \vec{g}$$

$$\sum \vec{F} = m \cdot \vec{a}$$

$$\vec{F}_g = m \vec{g}$$

Ağırlık = $m \cdot g$



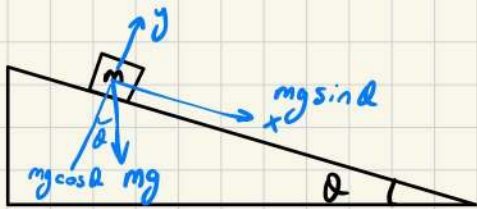
$$\left. \begin{aligned} T_{2x} &= T_2 \cos 53^\circ \\ T_{2y} &= T_2 \cdot \sin 53^\circ \\ T_{1x} &= T_1 \cdot \cos 37^\circ \\ T_{1y} &= T_1 \cdot \sin 37^\circ \end{aligned} \right\} =$$

$$T_1 = \frac{T_2 \sin 37^\circ}{\cos 37^\circ} = \frac{3 T_2}{4}$$

$$\frac{4}{5} (T_1 + T_2) = 100$$
$$\frac{3 T_2 + T_2}{4} = 100$$

$$\frac{4 T_2}{4} = 100$$

$$\frac{500}{7} = T_2$$
$$\frac{1500}{28} = T_1$$

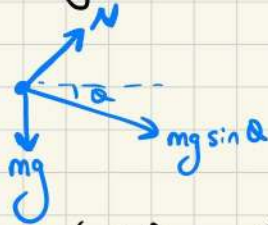


b) eşik düzlemde "d" kadar yol katıdıkse hızı ne olur.

$$v^2 = v_0^2 + 2a_x \cdot (x_s - x_0)$$

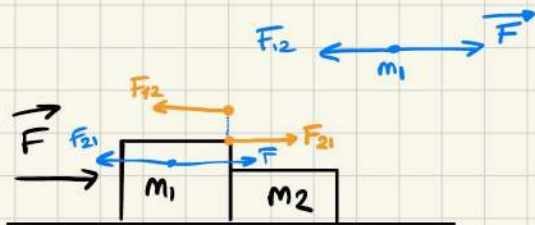
$$v = \sqrt{2g \sin \theta d}$$

a) bloğun ivmesi



$$mg \sin \theta = m \cdot a_x$$

$$g \sin \theta = a_x$$



a) sistem ivme

$$\sum F_x = m \cdot a_x$$

$$F = (m_1 + m_2) a_x$$

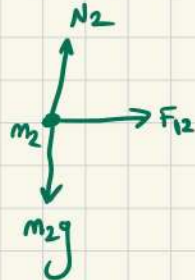
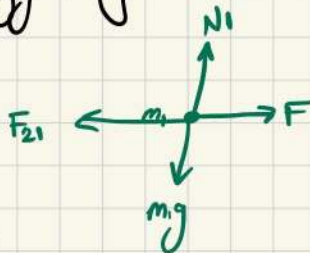
b) 1. 'nin 2. 'ye uyguladığı kuvvet?

$$\frac{F - F_{21}}{m_1} = a_1 = a_x$$

$$\frac{F}{m_1 + m_2} = \frac{F - F_{21}}{m_1}$$

$$F_{m1} = F_{m1} + F_{m2} - F_{21} m_1 - F_{21} m_2$$

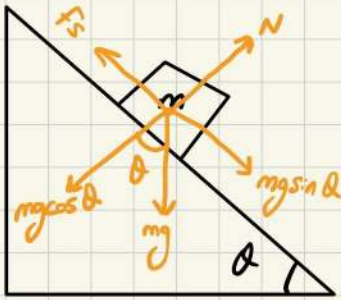
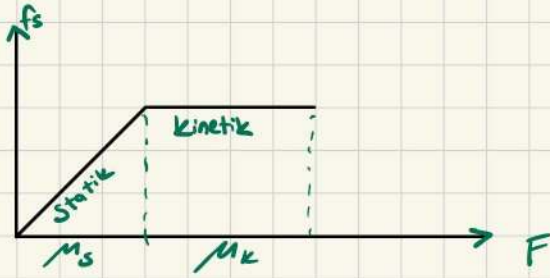
$$F_{21} = \left(\frac{m_2}{m_1 + m_2} \right) \cdot F \rightarrow$$



$$\frac{F_{12}}{m_2} = \frac{F}{m_1 + m_2}$$

$$F_{12} = -F \left(\frac{m_2}{m_1 + m_2} \right)$$

Sürtünme Kuvveti



Ör/ $N = mg \cos \alpha$

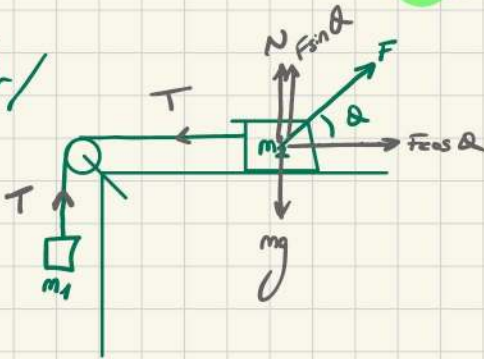
$$f_s = \mu_s N = \mu_s mg \cos \alpha$$

Eğer $\Sigma F_x = 0$ ise hareket yoktur.

$$\mu_s = \frac{\sin \alpha}{\cos \alpha} = \tan \alpha$$

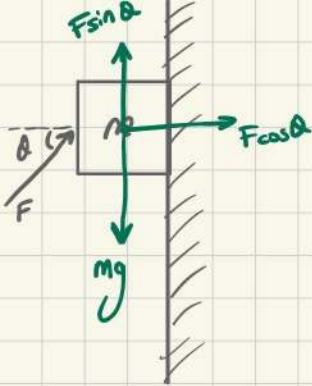
$$\mu_s = \tan \alpha_c$$

Ör/



ivmeleri gösteriniz.

$$\frac{T - m_1 g}{m_1} = \frac{F \cos \alpha - T - \mu(m_2 g - F \sin \alpha)}{m_2}$$



$$m = 3 \text{ kg} \quad \alpha = 50^\circ$$

Statik sürtünme $\mu = 0,25$ ise
cisim dengede tutmak muhtemel F nedir?

$$N = F \cos \alpha$$

$$f_s = \mu F \cos \alpha$$

$$f_s = F \sin \alpha + mg = \mu F \cos \alpha$$

$$F = \frac{mg}{\mu \cos \alpha + \sin \alpha} = 31.72 \text{ N}$$

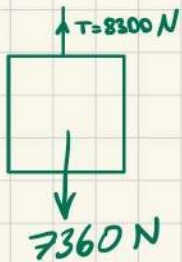
1. Durum

$$F \sin \alpha = mg + \mu F \cos \alpha$$

$$F = 48.56 \text{ N}$$

Ör/

75 kilo insan tutulmak isteniyor.

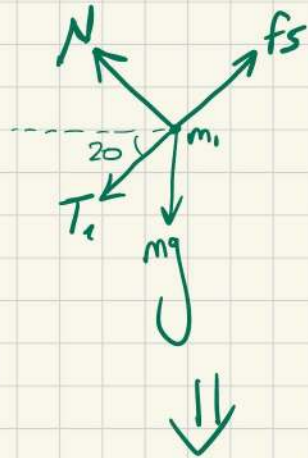
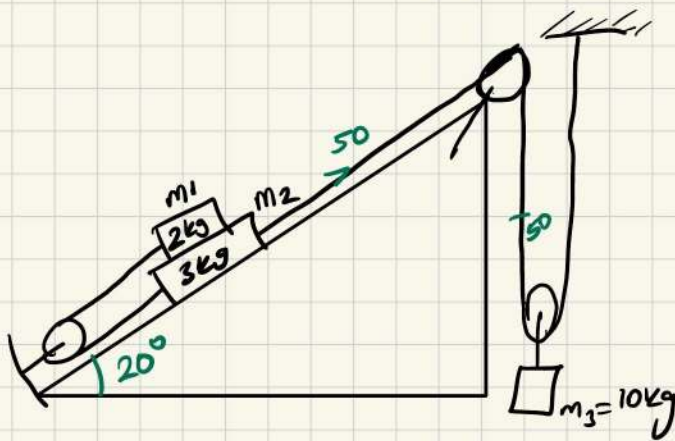


$$\frac{940}{750} = 1,25 \text{ m/s}^2$$

$$+ 9,8$$

$$\hline 11,05 \text{ m/s}^2$$

$$11,05 \cdot 75 = 82,8 \text{ N}$$



$$m_1 a = T_1 + m_1 g \sin 20^\circ - \mu_1 m_1 g \cos 20^\circ$$

$$2a - 2 = T_1$$

$$m_2 a = T_2 - m_2 g \sin 20^\circ - F_{s2} - F_{s1} - T_1$$

$$3a = T_2 - 7.5 - T_1$$

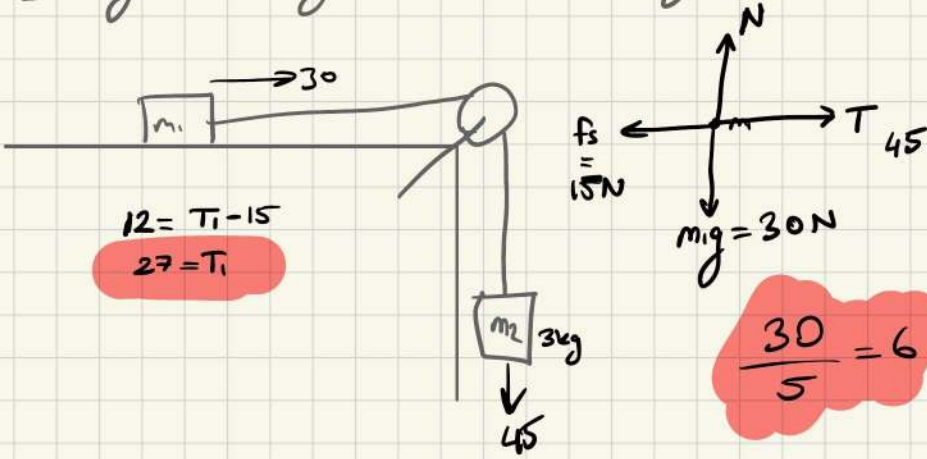
$$F_{s2} = \mu_2 N_2$$

$$m_3 g - 2T_2 = m_3 \cdot \frac{a}{2}$$

Şekilde yukarıya doğru $A = 5 \text{ m/s}^2$ 'lik ivmeyle hareket etmekte olan asansörün içinde serbest bırakılan sistemde yataydaki kinetik sürtünme katsayısı 0.5

$$m_1 = 2 \text{ kg}$$

$m_2 = 3 \text{ kg}$ old. göre blokların ivmesi bulunuz.



$$\frac{30}{5} = 6$$

$$45 - T_1 = 18$$

$$T_1 = 27$$

2 koşu sporcusu durgun halden aynı anda 100 metre koşuya başlıyor.
Her iki sporcu 10.2 sn de koşuyu tamamlıyor. 1. koşucun ivmesi = 2 sn
2. koşucunun ivme = 3 sn
ve sonra sbt hızda ilerliyorlar. Koşucuların hızları nedir?

$$x_{s1} = \frac{1}{2} a_1 t^2 = 2a_1$$

$$x_{s2} = \frac{1}{2} a_2 t^2 = 4,5a_2$$

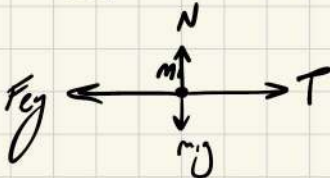
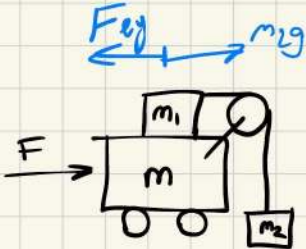
$$x_1 = v_{0x}(8,2) + 2a_1 = 100$$

$$x_2 = v_{0y}(7,2) + 4,5a_2 = 100$$

Blokların arabaya göre hareketsiz kalabilmesi için uygulanan $F = ?$

(Tüm yüzeyler, tekerlekler, makara sürtünmesizdir)

(iptek gerilme m_1 kütlesi meydana getirir)



$$F = (m + m_1 + m_2) a$$

$$a = \frac{F}{m + m_1 + m_2}$$

$$Fey = \frac{F \cdot m_1}{m + m_1 + m_2} = T$$

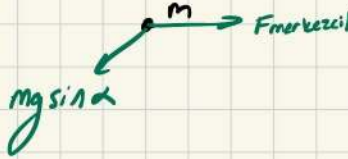
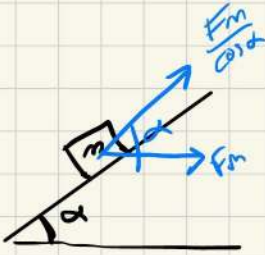
$$m_2 g =$$

Bir mühendis arabaların eğimli bir otayol istiyor.

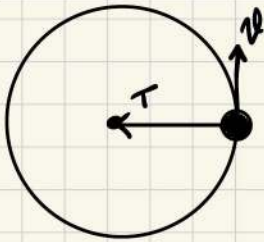
sürtünmeye güvenmeksizin savrulmadan dönebileceği

$$v = 13.4 \text{ m/s'lik hız}$$

$$\text{virajın yarıçapı} = 50 \text{ m}$$



Dairesel Hareket



$$T = \frac{m \cdot v^2}{r}$$

merteziil kuvvet
(F_r) = $m \cdot a_r = \frac{m \cdot v^2}{r}$

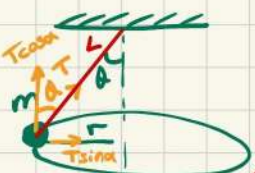
$$T_2 = mg \left(\frac{v^2}{r \cdot g} - 1 \right)$$

$$T_1 = mg \cdot \left(1 + \frac{v^2}{r \cdot g} \right)$$

$$T - mg \cos \alpha = \frac{m \cdot v^2}{r}$$

$$P_3 = mg \left(\cos \alpha + \frac{v^2}{r \cdot g} \right)$$

$$v = \sqrt{r \cdot g}$$



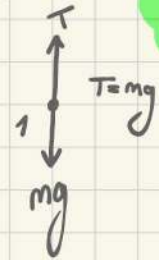
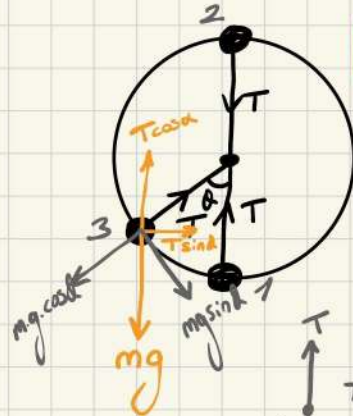
$$T \sin \alpha = \frac{m \cdot v^2}{r}$$

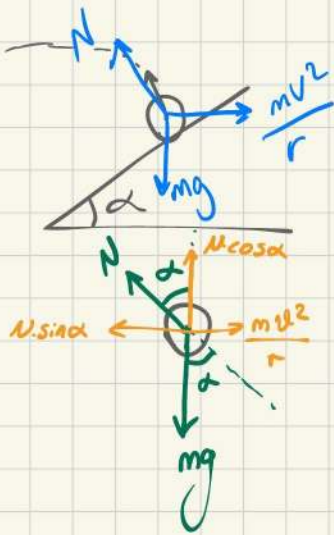
$$T \cos \alpha = mg$$

$$\tan \alpha = \frac{v^2}{r \cdot g}$$

$$v = \sqrt{r \cdot g \cdot \tan \alpha}$$

$$v \sqrt{L \cdot g \cdot \sin \alpha \cdot \tan \alpha}$$





Tekerlekler yol arasında sürtünmesiz ortamda ilerlemesi için açısı = ?

$$v = 13.4 \text{ m/s} \quad r = 35 \text{ m}$$

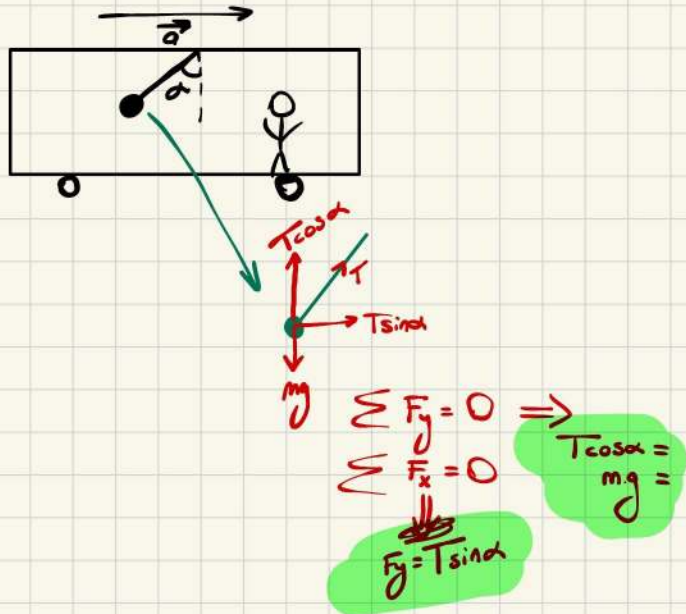
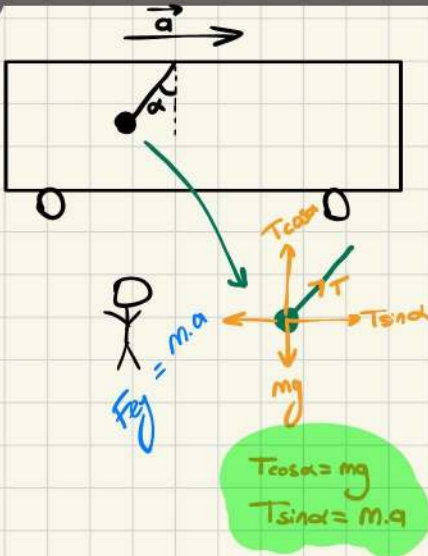
$$\frac{N \sin \alpha = \frac{mv^2}{r}}{N \cos \alpha = mg} = \tan \alpha = \frac{v^2}{r \cdot g}$$

$$v = \sqrt{r \cdot g \cdot \tan \alpha}$$

$$13.4 = \sqrt{35 \cdot 9.8 \cdot \tan \alpha}$$

$$\alpha = 27.6^\circ$$

Eylemlili Referans Sisteminde Hareket



Direnç Ortamında Hareket

$$v_0 = 0$$

$R = \text{direnç kuvveti} \sim v, v^2$

$$R = -k \cdot v$$



$$\Sigma F = m \cdot a$$

$$mg - kv = m \cdot \frac{dv}{dt}$$

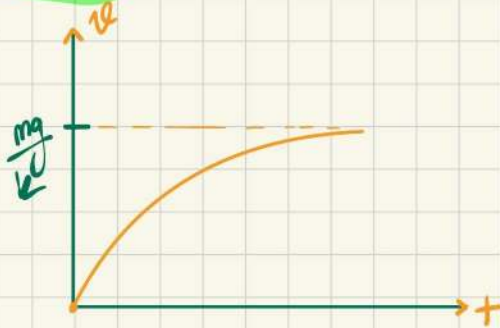
$$\frac{dv}{dt} = g - \frac{k}{m}v$$

$$\int \frac{dv}{g - \frac{k}{m}v} = \int_0^t dt$$

$$v(t) = \frac{mg}{k} (1 - e^{-\frac{k}{m}t})$$

$$v(t) = \frac{mg}{k} \left(1 - \frac{1}{e^{\frac{k}{m}t}} \right)$$

$$v_{\text{limit}} = \frac{m \cdot g}{k}$$



Cismin hızının $\lim$ hızın %63.2 değerine ulaşması için gereken süreye **zaman sabiti** denir.

2gr'lik cisim yağ içerisinde serbest bırakılıyor. Cisim yağdan bir direnç hissediyor.
 $v_{\lim} = 5 \text{ m/s}$

a) zaman sabiti

b) hızın, limit hızın %90'a ulaşması için geçen süre

$$\frac{m \cdot g}{k} = v_{\lim}$$

$$v = v_{\lim} (1 - e^{-\frac{t}{\tau}})$$

$$t = -5.1 \times 10^{-3} \ln(0.1) = 0.01 \text{ sn}$$

$$\frac{2 \cdot (980)}{5} = 392 \text{ gram/saniye} = k$$

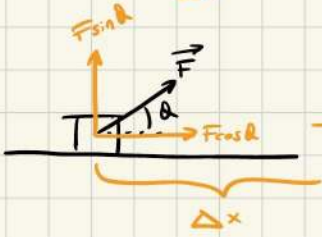
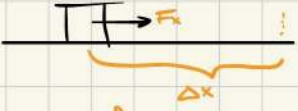
$$k = \frac{m}{\tau(s)} \rightarrow \text{g/s}$$

$$\tau = \frac{2}{392} \approx 5.1 \text{ m/s}$$

İş ve Enerji

Sabit Kuvvetin Yaptığı İş

$$W \equiv F_x \cdot \Delta x$$



$$W = F \cos \theta \cdot \Delta x$$

Conclusion

* Kuvvet, yer değiştirme ile aynı yönde iş yapmalıdır.

Skaler çarpım gereğidir.

$$W = F \cdot \Delta r \cdot \cos \theta$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

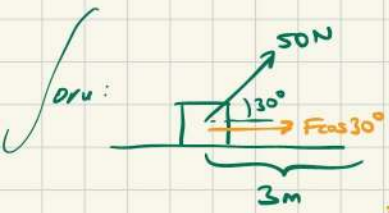
$$W = \vec{F} \cdot \Delta \vec{r}$$

Skaler çarpım

$$W = \vec{F} \cdot \Delta \vec{r}$$

$$N \cdot m = kg \frac{m^2}{s^2}$$

1 joule = Bir cisme 1 Newton kuvvet uygulanması ile 1 metre yer değiştirmesi sonucu kazandığı enerjiye denir.



W = ?

$$50 \cdot \cos 30^\circ \cdot 3 = \frac{150\sqrt{3}}{2} = 75\sqrt{3}$$

Değişen Kuvvetin Yapığı İş



$$W \approx \Delta x_1 \cdot F_1 + \Delta x_2 \cdot F_2 + \dots$$

$$W \approx \sum_{i=1}^6 F_i \Delta x_i$$

$$W = \lim_{\Delta x_i \rightarrow 0} \sum_{i=1}^{\infty} F_i \Delta x_i$$

$$\vec{F} = F_x \cdot \hat{i} + F_y \cdot \hat{j} + F_z \cdot \hat{k}$$

$$d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

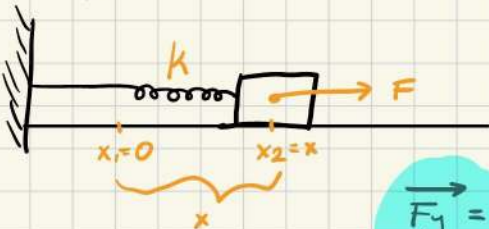


$$\vec{F} \cdot d\vec{r} = F_x \cdot dx + F_y \cdot dy + F_z \cdot dz$$

Vektörel Çarpım

$$W = \int_{x_i}^{x_s} F_x \cdot dx$$

Yay Kuvvetinin Yapığı İş



$$F \sim x$$

$$\vec{F} = F_x \cdot \hat{i}$$

$$\vec{F}_y = -k \cdot x \cdot \hat{i}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

$$W = -k \int_0^x x \cdot dx$$

$$W = -\frac{1}{2} k x^2$$

İş - Kinetik Enerji Teoremi

$$W = \int \sum \vec{F} \cdot d\vec{r}$$

$$\left. \begin{aligned} \sum \vec{F} &= m \cdot \vec{a} = m \cdot \frac{d\vec{v}}{dt} \\ W &= \int m \cdot \frac{d\vec{v}}{dt} \cdot d\vec{r} \end{aligned} \right\}$$

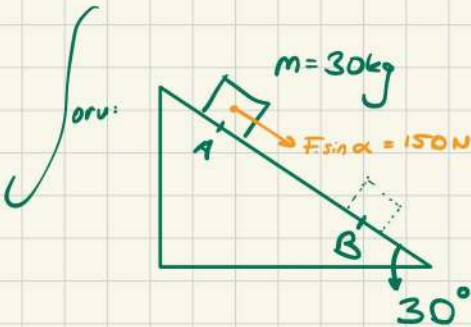
$$\left. \begin{aligned} W &= m \int d\vec{v} \cdot \frac{d\vec{r}}{dt} \\ W &= m \int_{v_i}^{v_s} v \cdot dv \end{aligned} \right\}$$

$$W = \frac{m \cdot v^2}{2} \Big|_{v_i}^{v_s}$$

$$W = \frac{1}{2} m \cdot v_s^2 - \frac{1}{2} m \cdot v_i^2$$

$$K_{(\text{kinetik enerji})} = \frac{1}{2} m v^2$$

$$W = \Delta K = K_s - K_i$$

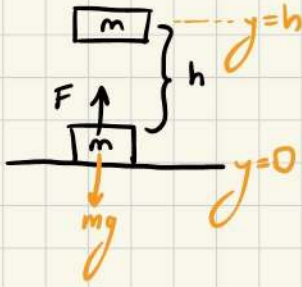


$$150 \cdot 15 = \frac{30 \cdot 2^2}{2} - \frac{30 \cdot v_B^2}{2}$$

$$150 = \frac{15}{30(4 - v_B^2)}$$

$$12.08 = \sqrt{146} = v_B$$

Kütle Çekim Potansiyel Enerjisi:



$$W = \int \Sigma \vec{F} \cdot d\vec{r}$$

$$\Sigma \vec{F} = F\hat{j} = mg\hat{j}$$
$$d\vec{r} = dy\hat{j}$$

$$W = mgh$$

$$U = mgh$$

(poten.)

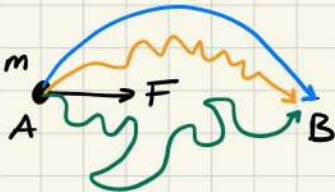
Elastik Potansiyel Enerji



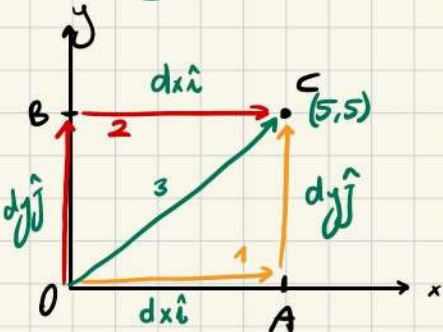
$$W = k \int_0^x x \, dx = \frac{1}{2} kx^2$$

$$U_y = \frac{1}{2} kx^2$$

Korunumlu Kuvvetler



Yapılan iş, yola bağlı değilse, yeri gidilen yoldan bağımsızsa **Korunumlu Kuvvet** denir.



$$\vec{F} = 2y\hat{i} + x^2\hat{j}$$

Bu cisim üç farklı yolla gidiyor. Korunumlu kuvveti araştırdık.



$$W_{OA} = \int \vec{F} \cdot d\vec{r}$$

$$W_{OA} = \int_0^5 (2y\hat{i} + x^2\hat{j}) \cdot (dx\hat{i}) = 0 \rightarrow A'ya \text{ giderken is yapilmedi.}$$

$$W_{OA} = 2 \int_0^5 y \, dx = 0$$

$$W_{AC} = \int_0^5 (2y\hat{i} + x^2\hat{j}) \cdot dy\hat{i} = x^2 \int_0^5 dy = x^2 (y_0^5)$$

$$W_{AC} = 5^2 \cdot 5 = 125 \text{ Joule}$$

$$W_{OB} = \int (2y\hat{i} + x^2\hat{j}) \cdot dy\hat{j} = \int x^2 dy = 0$$

$$W_{BC} = \int (2y\hat{i} + x^2\hat{j}) \cdot dx\hat{i} = 2y \int_0^5 dx = 2 \cdot 5 \cdot 5 = 50 \text{ Joule}$$

Korunumlu Kuvvetler bir sistemin potansiyel enerji fonksiyonlarından türetilirler.

$$\Delta K + \Delta U = 0$$

$$W = \int \vec{F} \cdot d\vec{r} = \Delta K = -\Delta U(x, y, z)$$

$$\vec{F} \cdot d\vec{r} = -dU(x, y, z)$$



$$F_x dx + F_y dy + F_z dz = -dU(x, y, z)$$

$$\vec{F} = -\vec{\nabla} U$$

Örnek: $U(x, y) = 3x^3y - 7x$
pot. enj. fonksiyonu

sisteme etki eden kuvveti bulunuz.

$$\vec{F} = -(9x^2y - 7)\hat{i} - (3x^3)\hat{j}$$

$$\vec{F}_x = -9x^2y + 7$$

$$\vec{F}_y = -3x^3$$

Soru: Bir cisim $F = 4x\hat{i} + 3y\hat{j}$ etkisi altında
 $x=5$ noktasına hareket ediyor. İşini bulunuz.

$$W = \int \vec{F} \cdot d\vec{r}$$

$$\vec{F} = 4x\hat{i} + 3y\hat{j}$$

$$d\vec{r} = dx\hat{i}$$

$$W = \int_0^5 4x dx = 2x^2 \Big|_0^5 = 50 \text{ Joule}$$

Soru: 3 kg kütleli cismin hız vektörü $\vec{v} = (6\hat{i} - 1\hat{j}) \text{ m/s}$ ile hareket ediyor.

55.5 J

a) Cismin K = ?

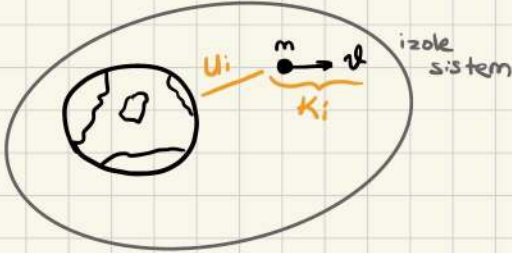
$$\frac{mv^2}{2} \rightarrow \frac{1}{2} \cdot 3 \cdot \underbrace{[(6\hat{i} - 1\hat{j}) \cdot (6\hat{i} - 1\hat{j})]}_{37} = \frac{101}{2} = 50.5 \text{ Joule}$$

b) Cismin hızı $(8\hat{i} + 4\hat{j}) \text{ m/s}$ değerine değişirse ne kadar iş yapılmış olur?

$$\frac{3 \cdot (8\hat{i} + 4\hat{j})^2}{2} = 120 \text{ Joule} \quad 120 - 55.5 = 64.5$$

Enerji Korunumu

İzole Olmuş Sistem

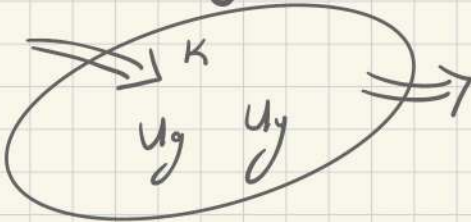


$$\sum E_i = \sum E_s$$

$$K_i + U_i = K_s + U_s$$

$$\Delta K + \Delta U_g + \Delta U_y = 0$$

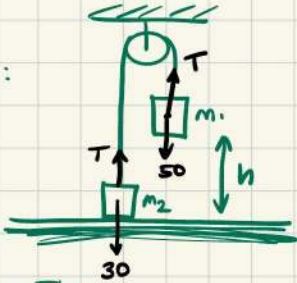
İzole Olmayan Sistem



$$W = \int \vec{F} \cdot d\vec{r}$$

$$W_s = - \int f ds$$

Soru:



$$m_1 = 5 \text{ kg}$$

$$m_2 = 3 \text{ kg}$$

$$h = 4 \text{ m}$$

m_1, m_2 aynı yükseklikte hızları ne olur?

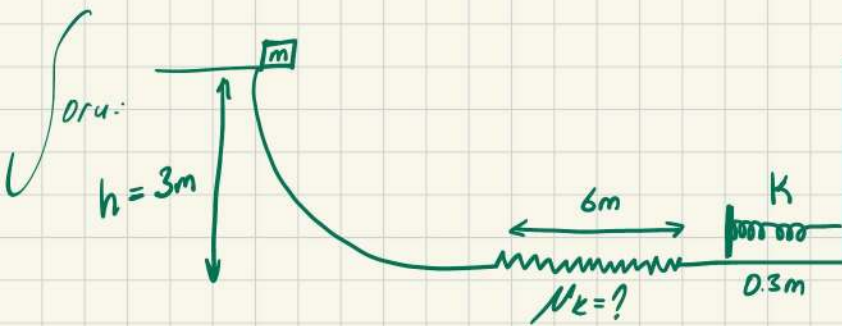
$$E_1 = E_s = 50 \cdot 4 = 200$$

$$v_1 = v_2 = v$$

$$200 = 80 \cdot 2 + \frac{m_1 v_1^2}{2} + \frac{m_2 v_2^2}{2}$$

$$40 = 4v^2$$

$$\sqrt{10} = v$$



$$m = 10\text{kg}$$

$$K = 2250\text{ N/m}$$

$$E = 300\text{J}$$

$$E_{\text{springe}} = 600 \cdot \mu_k$$

$$(U_{fs} - U_{fi}) + (U_{gs} - U_{gi}) + (K_s - K_i) = 0$$

$$300 - 600\mu_k = \frac{1}{2} K \cdot x^2$$

$$101.25$$

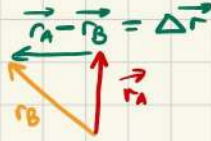
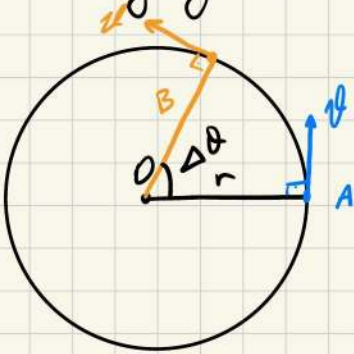
$$\mu_k = 0.33$$

$$300 - 600\mu_k = \frac{1}{2} \cdot 2250 \cdot \frac{9}{10}$$

$$\mu_k = 0.33$$

Düzgün Dairesel Hareket

- Hızın büyüklüğü sabittir.



$$|\vec{v}_A| = |\vec{v}_B| = v$$

$$|\vec{r}_A| = |\vec{r}_B| = r$$

$$\frac{|\Delta \vec{r}|}{r} = \frac{|\Delta \vec{v}|}{v}$$

$$|\Delta \vec{v}| \stackrel{||}{=} \frac{v \cdot |\Delta \vec{r}|}{r}$$

$$|\vec{a}_{ort}| = \frac{|\Delta \vec{v}|}{\Delta t}$$

$$|\vec{a}_{ort}| = \frac{v}{r} \cdot \frac{|\Delta \vec{r}|}{\Delta t}$$

$$a = \frac{v^2}{r}$$

merkezcil ivme büyüklüğü



1) Hızın büyüklüğünün değişiminden dolayı sahip olduğu ivme

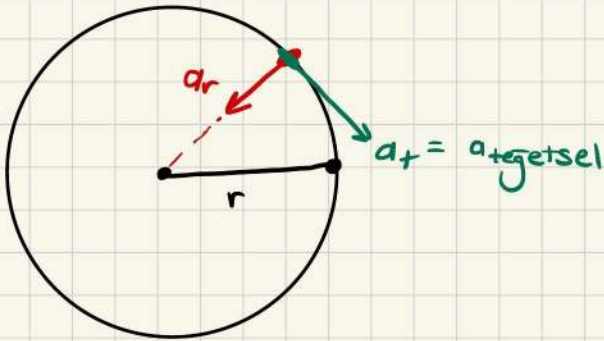
$$a_1 = \frac{dv}{dt}$$

2) Hızın yönünün değişmesinden dolayı sahip olduğu ivme

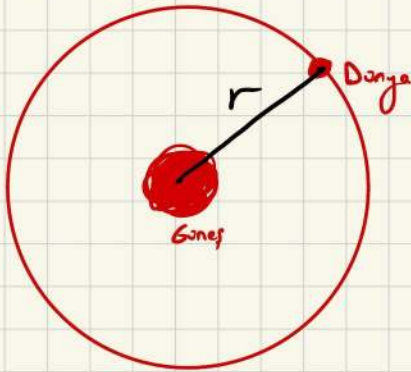
$$a_{\text{toplam}} = \vec{a}_1 + \vec{a}_2$$

$$a_2 = \frac{v^2}{r}$$

Dairesel Hareket



Dünyanın Güneş etrafında dönmesi sonucu sahip olduğu merkezî ivme nedir?

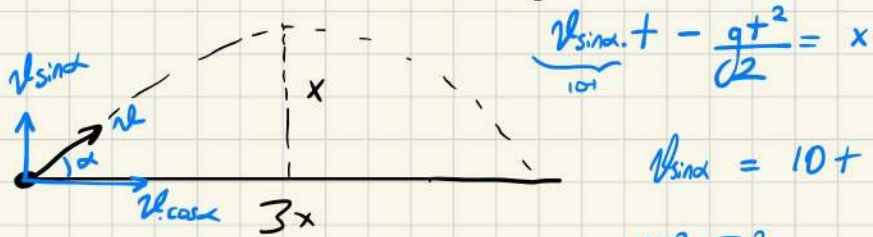


$$a = \frac{v^2}{r}$$

$$2\pi r = v \cdot t$$

$$\frac{\left(\frac{2\pi r}{t}\right)^2}{r} = \frac{4\pi^2 r}{t^2} = a$$

Bir taş yerden belli bir açıyla fırlatılınca



$$\frac{v \sin \alpha \cdot t - \frac{g t^2}{2}}{10^1} = x$$

$$v \sin \alpha = 10 +$$

$$\frac{10 + 2 - 5 + 2}{5 + 2} = x$$

$$\tan \alpha = \frac{10 \cdot 2}{2 \cdot 18} = \frac{10}{9}$$

$$v \cos \alpha \cdot 2 = 18$$

$$\tan \alpha = \frac{4}{3}$$

$$v \cos \alpha = \frac{15}{2}$$

$$\alpha = 53^\circ$$

Başlangıçta bir parçacığın $\vec{a} = 3\hat{j} \text{ m/s}^2$

$v_0 = 5\hat{i} \text{ m/s}$ olarak veriliyor.

x konum vektörü = ?

hız vektörü = ?

$t=2$ için koordinat = ?

$t=2$ için hız = ?

$$x = 0 + 5\hat{i}t + \frac{3\hat{j}t^2}{2}$$

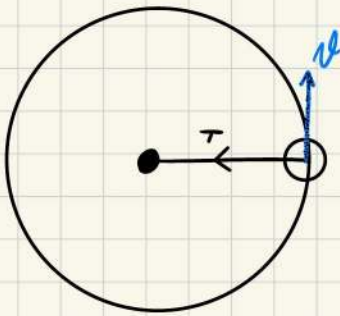
$$x = 5\hat{i} + \frac{3\hat{j}}{2}$$

$$v = 5\hat{i} + 3\hat{j}$$

$$(10, 6)$$

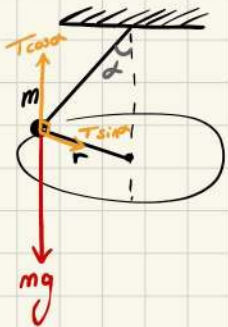
$$(5\hat{i} + 6\hat{j}) = \text{hız}$$

$$25 + 36 = \sqrt{71} = \text{hız}$$



$$T = \frac{m \cdot v^2}{r}$$

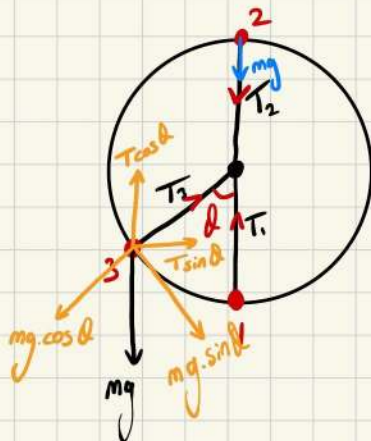
$$F_r = m \cdot a_r = \frac{m \cdot v^2}{r}$$



$$T \sin \alpha = \frac{m \cdot v^2}{r}$$

$$T \cos \alpha = mg$$

$$\Rightarrow v = \sqrt{r \cdot g \cdot \tan \alpha}$$

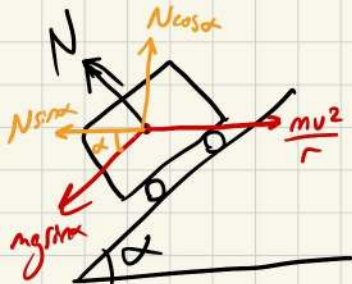


$$3-) T - mg \cos \alpha = \frac{m v^2}{r}$$

$$T_3 = mg \left(\cos \alpha + \frac{v^2}{rg} \right)$$

$$T_1 = mg \left(1 + \frac{v^2}{rg} \right) = mg + \frac{m v^2}{r}$$

$$T_2 = \left(\frac{m v^2}{r} - mg \right)$$



$$v = 13.4 \text{ m/s}$$

$$r = 35 \text{ m}$$

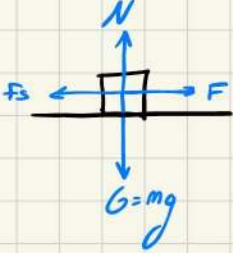
$$N \sin \alpha = \frac{mv^2}{r}$$

$$N \cos \alpha = mg$$

$$\tan \alpha = \frac{v^2}{r \cdot g}$$

Serbest Cisim Diyagramı

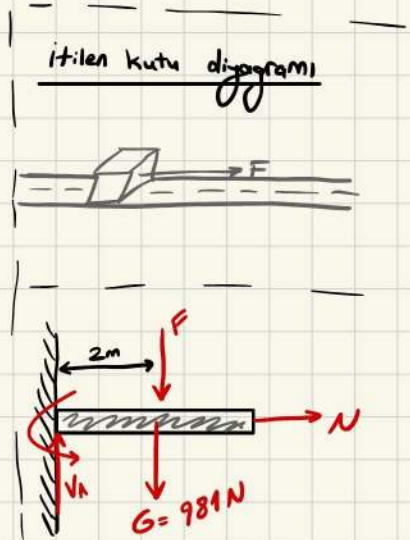
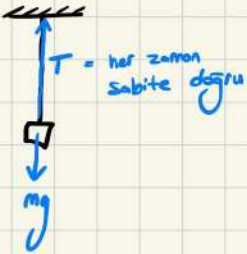
Üstünde cisme etki eden bütün kuvvetler ölçeksiz gösterilir.

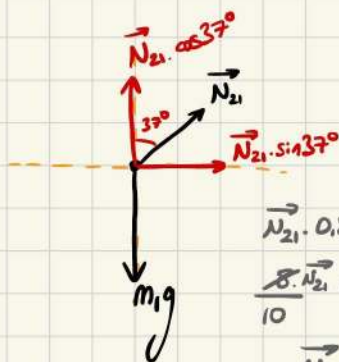
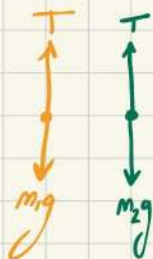
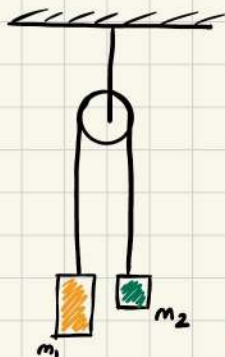
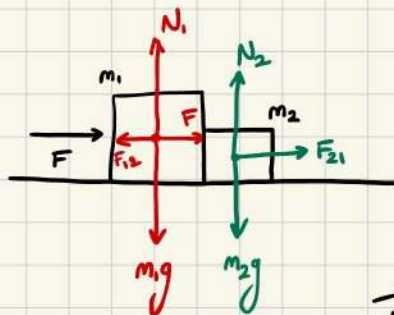
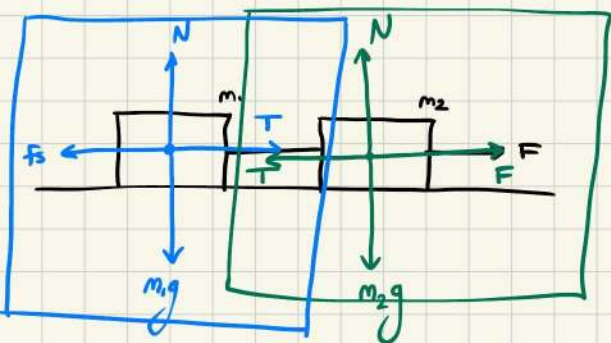
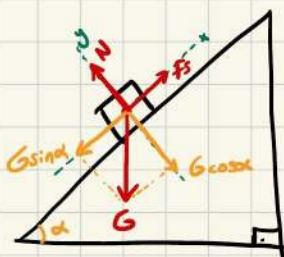
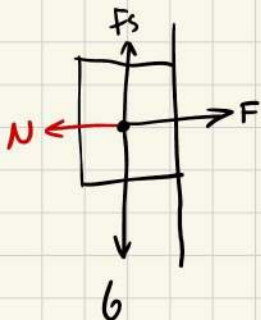


Taş binanın tepesinden bırakılıyor. Taşa etkiyen kuvvet diyagramı = ?



Tavandaki avize serbest cisim diyagramı





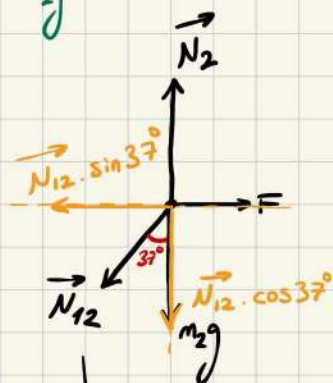
$$\vec{N}_{21} \cdot 0,8 = m_1 \cdot g$$

$$\frac{8 \cdot \vec{N}_{21}}{10} = 24,3$$

$$\vec{N}_{21} = 30 \text{ N}$$

$$\frac{30 \cdot \vec{N}_{21} \cdot 6}{10} = 2,4 \cdot a$$

$$\frac{30}{4} = \frac{3}{0,4} = \frac{18}{2,4} = a$$



$$F = \vec{N}_{12} \cdot \frac{6}{10} = 4 \cdot \frac{30}{4}$$

$$\frac{-30,6}{10} = 30$$

$$\frac{-18}{10} = 30$$

$$\boxed{F = 48}$$

Lineer Momentum ve Korunumu



$$\begin{aligned}\sum \vec{F}_1 &= \cancel{\vec{F}_{21}} + \vec{F}_{31} + \vec{F}_{41} \\ \sum \vec{F}_2 &= \vec{F}_{32} + \cancel{\vec{F}_{12}} + \vec{F}_{42}\end{aligned}$$

* Bir kuvvetsiz düzlemde kütlelerin birbirine uyguladığı toplam kuvvet "0" dir.

$$\sum \vec{F}_1 + \sum \vec{F}_2 + \sum \vec{F}_3 = 0$$

$$m_1 \vec{a}_1 + m_2 \vec{a}_2 + m_3 \vec{a}_3 = 0$$

$$m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt} + m_3 \frac{d\vec{v}_3}{dt} = 0$$

$$\frac{d}{dt} (m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3) = 0$$

$$m \cdot \vec{v} = \vec{P}$$

$$\vec{P}_{top} = sbt$$

$$* a = \frac{dv}{dt}, \alpha = \frac{d\omega}{dt}$$

Kuvvetsiz sistemde $P_i = P_s$

* Dışarıdan kuvvet uygulanmıyorsa lineer momentum muhakkak esittir.

Soru: Sürtünmesiz düzlemde 50kg okçu 0,5kg kütlesi bir ok fırlatıyor. Okun hızı 50m/s

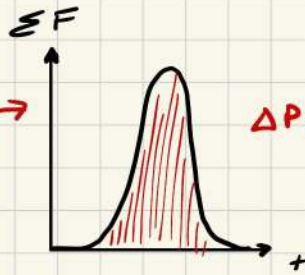
ise okçunun hızı=?

$$\begin{aligned}50 \cdot x &\leftarrow \rightarrow 0,5 \cdot 50 = 25 \text{ kg} \cdot \text{m/s} \\ x &= -0,5 \text{ m/s}\end{aligned}$$

$$\Sigma \vec{F} = m \cdot \frac{d\vec{v}}{dt}$$

$$\Sigma \vec{F} = \frac{d\vec{P}}{dt}$$

$$\int_{\vec{P}_i}^{\vec{P}_s} d\vec{P} = \int_0^+ \Sigma \vec{F} \cdot dt$$



$\Delta P = \text{İtme}$

$$F_{ort} = \frac{1}{\Delta t} \int_{t_1}^{t_2} \Sigma F \cdot dt$$

İtme (Impuls) =

anlık momentum değişimi

$$\Delta P = F_{ort} \cdot \Delta t$$

1500 kg otomobil duvara çarpıtılıyor. Otomobilin çarpışmadan önceki ve sonraki hızları

$$\vec{v}_i = -15 \hat{i} \text{ m/s}$$

$$\vec{v}_s = 2.6 \hat{i} \text{ m/s}$$

Eğer çarpışma 0.15 sn sürüyorsa çarpışma itmeği ve otomobile etki eden ortalama kuvveti bulunuz.

$$2) 26500 = \frac{15}{100} \cdot F_{ort}$$

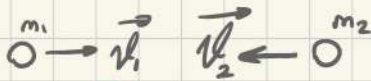
$$176000 \text{ N} = F_{ort}$$

$$1) \quad -22500 \hat{i} \text{ kg m/s} \longleftrightarrow 3900 \hat{i} \text{ kg m/s}$$

$$1500 \cdot -15 =$$

$$\Delta P = 26400 \hat{i}$$

Çarpışmalar



$$\sum \vec{P}_i = \sum \vec{P}_s$$

* Dışardan kuvvet bulunmıyor
tüm lineer durumlarda

$$m_1 v_{1i} + m_2 v_{2i} \dots = m_1 v_{1s} + m_2 v_{2s} \dots$$

eğer esnekse kinetik enerji korunacağından:

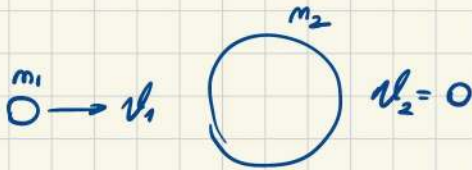
$$\frac{m_1 v_{1i}^2}{2} + \frac{m_2 v_{2i}^2}{2} + \dots = \frac{m_1 v_{1s}^2}{2} + \frac{m_2 v_{2s}^2}{2} + \dots$$

$$\vec{v}_{1i} + \vec{v}_{1s} = \vec{v}_{2i} + \vec{v}_{2s} = \dots$$

$$v_{1s} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2} \right) v_{2i}$$

$$v_{2s} = \left(\frac{m_2 - m_1}{m_2 + m_1} \right) v_{2i} + \left(\frac{2m_1}{m_1 + m_2} \right) v_{1i}$$

1. Durum



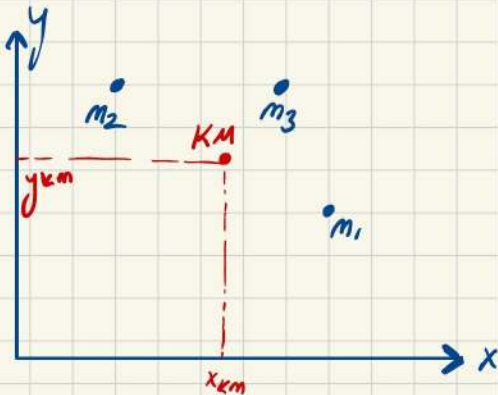
* m_2 ihmal edilebilecek kadar küçük

$$v_{1s} = -v_{1i}$$

$$v_{1s} = -v_{1i}$$

$$v_{2s} \approx 0$$

Kütle Merkezi

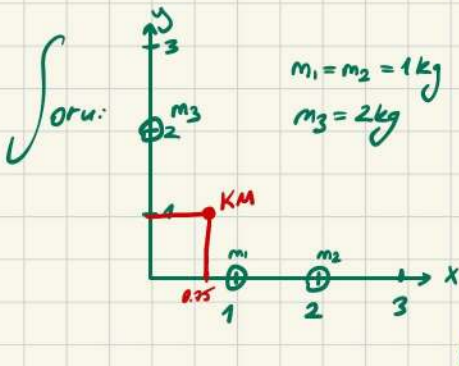


$$x_{km} = \frac{m_1 x_1 + m_2 x_2 + \dots}{m_1 + m_2 + m_3 + \dots}$$

$$x_{km} = \frac{1}{M} \sum_{i=1}^n m_i x_i$$

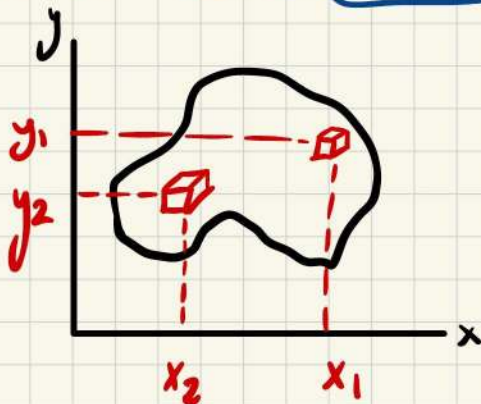
$$y_{km} = \frac{1}{M} \sum_{i=1}^n m_i y_i$$

$$\vec{r}_{km} = x_{km} \hat{i} + y_{km} \hat{j} + z_{km} \hat{k}$$



Sürekli Kütle Dağılımına Sahip Cisimlerin

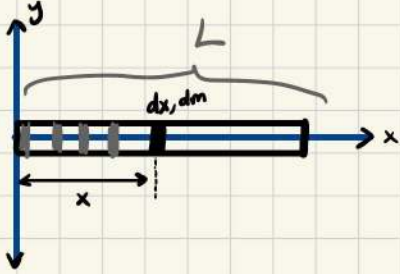
Kütle Merkezleri



$$x_{km} \approx \frac{1}{M} \sum_{i=1}^n \Delta m_i x_i$$

$$x_{km} \approx \frac{1}{M} \cdot \lim_{\Delta m_i \rightarrow 0} \sum_{i=1}^{\infty} \Delta m_i x_i$$

$$x_{km} = \frac{1}{M} \int x \, dm$$



$$\frac{L}{dx} \times \frac{M}{dm}$$

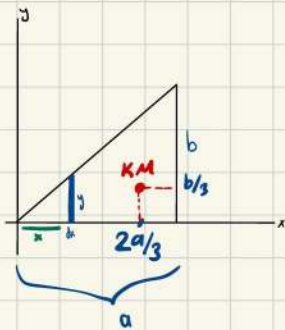
$$dm = \frac{M}{L} dx \quad \bigg| \quad dm = \lambda dx$$

←
gesamt masse längen

$$x_{km} = \frac{1}{M} \int x dm$$

$$x_{km} = \frac{1}{M} \int_0^L x \lambda dx$$

$$x_{km} = \frac{\lambda}{M} \cdot \frac{L^2}{2} = \frac{1}{2} L$$



$$dm = \sigma dA$$

$$\sigma = \frac{M}{A}; dA = y dx$$

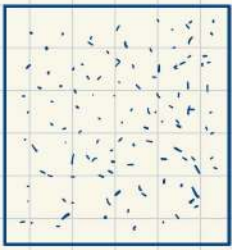
$$dm = \sigma y dx$$

$$x_{km} = \frac{1}{M} \int x dm$$

$$x_{km} = \frac{\sigma}{M} \int_0^a x y dx$$

$$x_{km} = \frac{2a}{3}$$

$$y_{km} = \frac{b}{3}$$



$$\vec{r}_{km} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

$$\vec{v}_{km} = \frac{d\vec{r}_{km}}{dt} = \frac{d}{dt} \left[\frac{1}{M} \sum_i m_i \vec{r}_i \right]$$

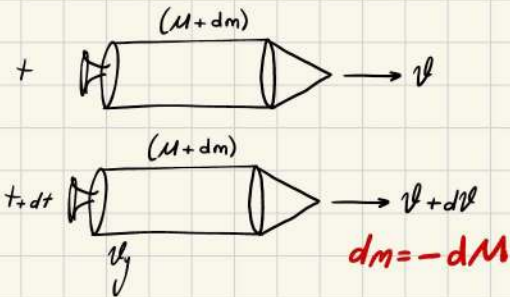
$$M \vec{v}_{km} = \sum_i m_i \cdot \vec{v}_i$$

$$\vec{a}_{km} = \frac{d\vec{v}_{km}}{dt} = \frac{d}{dt} \left[\frac{1}{M} \sum_i m_i \vec{v}_i \right]$$

$$M \cdot \vec{a} = \sum_i m_i \vec{a}_i$$

$$\vec{F} = \sum_i \vec{F}_i$$

Roket Hareketi



+ anında

$$(M+dm) v = M(v+dv) + dm(v-v_y)$$

$$v_s = v_i + v_y \ln \left(\frac{m_i}{m_s} \right)$$

$$T = M \cdot \frac{dv}{dt} \rightarrow T = \left| v_y \frac{dM}{dt} \right|$$

Roket hızı $3 \cdot 10^3 \text{ m/s}$

Bu anda roket ateşleniyor.

Roketin yakıtı $5 \cdot 10^3 \text{ m/s}$ 'lik hızla terk ediyor

Roketin kütlesi yarıya düşerse yere göre hız = ?

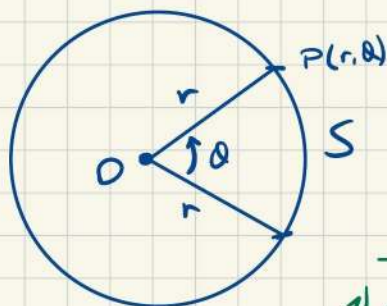
$$v_s = v_i + v_y \ln\left(\frac{m_i}{0.5m_i}\right)$$

$$(3 \cdot 10^3) + (5 \cdot 10^3) \cdot \ln 2 = 6.5 \cdot 10^3 \text{ m/s}$$

Roket yakıtı 50 kg/s oranında yanıyor itme kuvveti = ?

$$\left| v_y \frac{dm}{dt} \right| = T = 50 \cdot 5 \cdot 10^3 = 25 \cdot 10^4 \text{ N}$$

Katı Bir Cismin Sabit Ekseninde Dönmesi



Tom bir tur için

$$2\pi r = r \theta$$

$$2\pi = \theta \text{ (radyan)}$$

$$S = r \cdot \theta$$

$$\theta = \text{radyan}$$

1) Açısal Yer Değiştirme

$$\Delta \theta = \theta_s - \theta_i$$

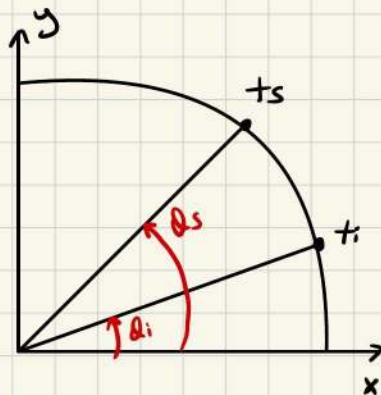
2) Ortalama Açısal Hız

$$\omega_{\text{ort}} = \frac{\Delta \theta}{\Delta t} \text{ (rad/s)}$$

3) Ani Açısal Hız

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t} = \frac{d\theta(t)}{dt}$$

türev operatörü



4) Ortalama Açısal İvme

$$\alpha_{\text{ort}} = \frac{\Delta \omega}{\Delta t} \text{ (rad/s}^2\text{)}$$

5) Ani Açısal İvme

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t} = \frac{d\omega(t)}{dt}$$

Sabit Açısal ivmeli: Dönme Hareketi

Dönme Hareketi

$$\omega_s = \omega_0 + \alpha t$$

$$\theta_s = \theta_0 + \omega_0 t + \frac{\alpha t^2}{2}$$

$$\omega_s^2 = \omega_0^2 + 2\alpha(\theta_s - \theta_0)$$

$$(\alpha = \text{sbt})$$

değilse;

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt}$$

Öteleme Hareketi

$$v_s = v_i + a \cdot t$$

$$x_s = x_i + v_i t + \frac{a t^2}{2}$$

$$v_s^2 = v_0^2 + 2a(x - x_0)$$

$$(a = \text{sbt})$$

değilse;

$$v = \frac{dx}{dt}$$

$$a = \frac{dv}{dt}$$

Soru: $\alpha = 3,5 \text{ rad/s}^2$

$$t_0 = 0 \Rightarrow \omega_0 = 2 \text{ rad/s}$$

a) $t = 2 \text{ sn}$ 'deki $\Delta\theta = ?$

$$\theta_s = \theta_0 + \omega_0 t + \frac{\alpha t^2}{2}$$

$$\Delta\theta = 4 + \frac{3,5 \cdot 4}{2} = 11 \text{ rad}$$

b) $t = 2 \text{ sn}$, kaç tur atar?

$$\frac{11}{2\pi} \text{ derece yani } \frac{180 \cdot (11)}{\pi} = 630^\circ$$

$$1,75 \text{ tur}$$

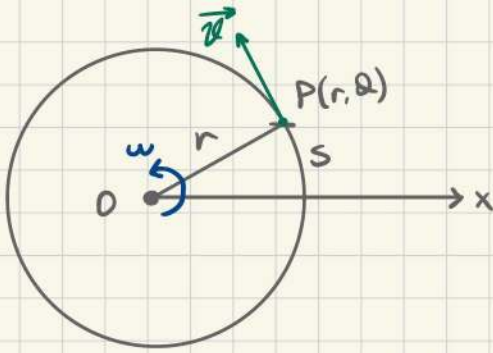
c) $t = 2 \text{ sn}$

açısal hız = ?

$$\omega_s = \omega_0 + \alpha t$$

$$\omega_s = 2 + 7$$

$$\omega_s = 9 \text{ rad/s}$$



$$s = r \theta$$

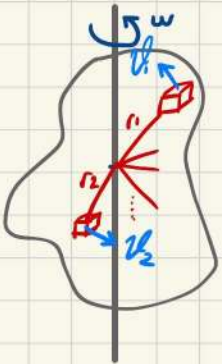
$$\frac{ds}{dt} = r \frac{d\theta}{dt}$$

$$v = r \omega$$

$$\frac{dv}{dt} = r \frac{d\omega}{dt}$$

$$a = \alpha r$$

Dönme Kinetik Enerjisi



$$K \approx \frac{1}{2} \Delta m_1 v_1^2 + \frac{1}{2} \Delta m_2 v_2^2 + \dots$$

$$K \approx \sum_i^N \frac{1}{2} \Delta m_i v_i^2$$

$$K \approx \frac{1}{2} \left(\sum_i^N \Delta m_i r_i^2 \right) \omega^2$$

$$K = \frac{1}{2} \left(\lim_{\Delta m_i \rightarrow 0} \sum_{i=1}^{\infty} \Delta m_i r_i^2 \right) \cdot \omega^2$$

- Sadece "ω" hepsinde aynı.

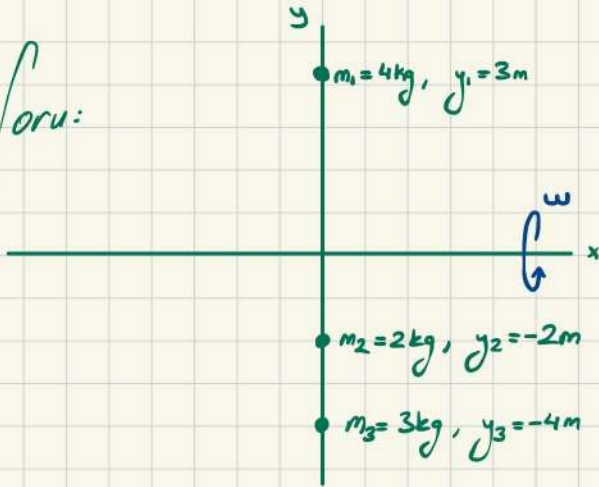
$$\int r^2 dm = I$$

- eylemsizlik momenti

$$K = \frac{1}{2} I \omega^2$$

$$\sum_i m_i r_i^2 = I$$

Soru:



$$\omega = 2 \text{ rad/s}$$

Sistemin kinetik enerjisi kaç J = ?

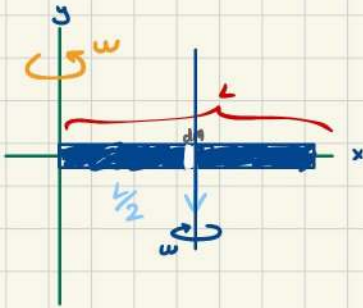
$$K = \frac{1}{2} I \omega^2$$

$$I = \sum_i m_i r_i^2 \text{ ise,}$$

$$\frac{4 \cdot 9}{36} + \frac{2 \cdot 4}{8} + \frac{3 \cdot 16}{48} = 92 \text{ kg} \cdot \text{m}^2$$

$$\frac{1}{2} 92 \cdot 2^2 = 184 \text{ J}$$

Soru:



$$I = \int x^2 dm$$

Eylemsizlik momenti nedir?

$$I = \int x^2 dm$$

$$I = \lambda \cdot \frac{L^3}{3}$$

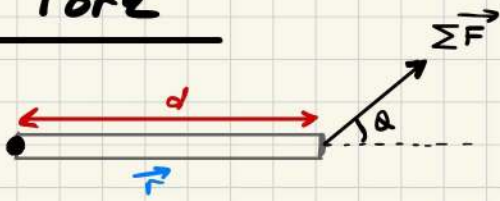
$$I = \frac{1}{3} M L^2$$

$$dm = \frac{M}{L} \cdot dx$$

$$I = \int_0^L x^2 \frac{M}{L} dx$$

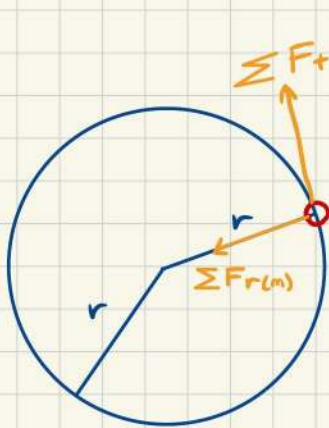
$$\frac{x^3}{3} \Big|_0^L = \frac{L^3}{3} \cdot \frac{M}{L} = \frac{M L^2}{3}$$

Tork



$$\tau = (\Sigma F \cdot \sin \alpha) \cdot d$$

$$\tau = \vec{r} \times \Sigma \vec{F}$$



merkezcil kuvvet tork üretmez.

$$\Sigma F_t = m \cdot \alpha_t$$

$$r \alpha = a_t$$

$$\Sigma \tau = (m \cdot r^2) \alpha$$

$$\Sigma \tau = I \alpha$$

Soru: m kitlesi çubuk yatayla paralelken serbest bırakılıyor. Aşağı açısal ivmeyi ve çubukun en ucunun çizgisel ivmesini bulunuz.



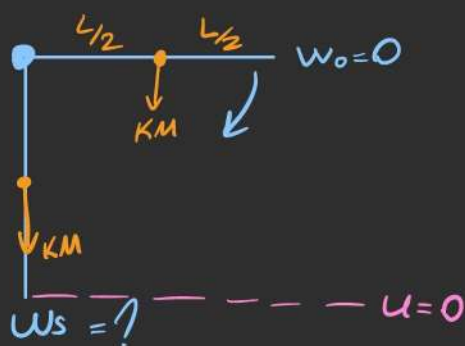
$$I = \frac{1}{3} m L^2$$

$$\tau = I \cdot \alpha = mg \cdot \frac{L}{2}$$

$$\frac{1}{3} m L^2 \cdot \alpha = mg \cdot \frac{L}{2}$$

$$\alpha = \frac{3g}{2L}$$

$$\frac{3g}{2L} \cdot L = \frac{3g}{2}$$



$$\Delta E = 0$$

$$mgL = K + mg \frac{L}{2}$$

$$\frac{mgL}{2} = K$$

$$\frac{mgL}{2} = \frac{1}{2} \omega_s^2$$

$$\frac{3mgK}{\text{or } L^2} = \omega_s^2$$

$$\sqrt{\frac{3g}{L}} = \omega_s$$

$$r: \omega = v$$

$$\frac{L}{2} \cdot \sqrt{\frac{3g}{L}} = v_{KM}$$

$$\sqrt{3gL} = v_p$$

Dönen katı cismin üzerindeki herhangi bir noktanın bir sabit eksenle yaptığı, açının zamanla değişimi

$$\theta = (5 + 10t + 2t^2) \text{ rad}$$

$t = 3$ sn için

açısız konum, hız, ivme

$$\omega = 10 + 4t \rightarrow 22$$

$$\alpha = 4 \rightarrow \alpha = 4$$

$$\theta = 53 \text{ rad}$$

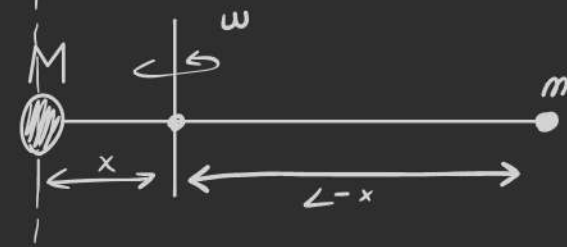
$$\alpha = (10 + 6t) \text{ rad/s}^2$$

$$10 + 6t = \frac{d\omega}{dt}$$

$$\omega(t) = (10t + 3t^2) \text{ rad/s}$$

$$\theta(t) = (5t^2 + t^3) \text{ rad}$$

$$\theta(4) = 144 \text{ rad}$$



$$I = M L^2 \quad \mu = \frac{Mm}{m+M}$$

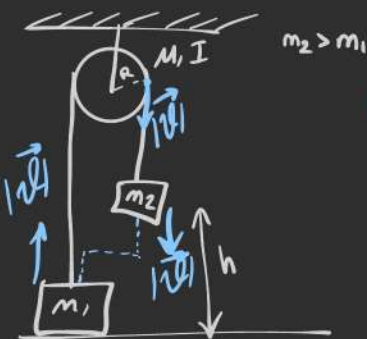
$$I = \sum_i m_i r_i^2$$

$$I = Mx^2 + m(L-x)^2$$

$$\frac{dI}{dx} = 2Mx + 2m(L-x)(-1)$$

$$Mx = mL - mx$$

$$x = \left(\frac{m}{m+M} \right) L$$



$$m_2 gh = \frac{m_2 gh}{2} + \frac{m_1 gh}{2} + \frac{m_1 v^2}{2} + \frac{m_2 v^2}{2} + \frac{I \omega^2}{2}$$

$$r \cdot \omega = v$$

$$\sqrt{\frac{(m_2 - m_1) gh}{m_1 + m_2 + \frac{I}{R^2}}} = v$$

$$840 = 320 J \quad \frac{20 \cdot 40.9}{2}$$

$$320 + 180 = 500 J = \frac{20}{25} \cdot \frac{40.9^2}{2}$$

$$v = 5$$

$$40.9 = 48.12$$

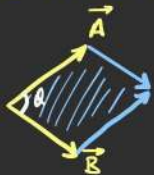
$$\vec{F}_1 \times \vec{r}_1 =$$

$$\begin{array}{ccc} i & j & k \\ 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{array} \quad \begin{array}{ccc} i & j & k \\ 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{array}$$

Açılma Momentum

$$\vec{C} = \vec{A} \times \vec{B}$$

$$C = A \cdot B \cdot \sin \theta$$

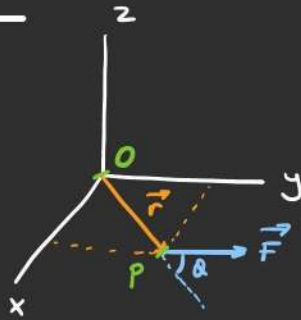


$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\frac{d}{dt} (\vec{A} \times \vec{B}) = \frac{d\vec{A}}{dt} \times \vec{B} + \vec{A} \times \frac{d\vec{B}}{dt}$$



$$\begin{aligned} \vec{r} &= x\hat{i} + y\hat{j} + z\hat{k} \\ \vec{p} &= p_x\hat{i} + p_y\hat{j} + p_z\hat{k} \\ L_x &= (y p_z - z p_y) \\ L_y &= (z p_x - x p_z) \\ L_z &= (x p_y - y p_x) \end{aligned}$$



$$\sum \vec{\tau} = \vec{r} \times \vec{F}$$

$$\sum \vec{\tau} = \frac{d}{dt} (\vec{r} \times \vec{p})$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

Soru

$$\vec{F} = (2\hat{i} + 3\hat{j}) \text{ N}$$

$$\vec{r} = (4\hat{i} + 5\hat{j}) \text{ m}$$

a) Net torque

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ 4 & 5 & 0 \end{vmatrix} = \hat{i}(3 \cdot 0 - 0 \cdot 5) - \hat{j}(2 \cdot 0 - 0 \cdot 4) + \hat{k}(2 \cdot 5 - 3 \cdot 4) = 10\hat{k} - 12\hat{k} = -2\hat{k}$$

2k

Noktasal Parçacıklar ise

$$\vec{L}_{top} = \sum \vec{L}_i$$

Rijit Cisim Açılma Momentumu



$$\vec{L}_i = \vec{r}_i \times \vec{p}_i$$

$$L_i = r_i \cdot \Delta m_i \cdot v_i = r_i \Delta m_i \cdot r_i \cdot \omega$$

$$L_{top} = \left(\sum r_i \Delta m_i \right) \omega$$

$$L_{top} = \left(\lim_{\Delta m_i \rightarrow 0} \sum r_i^2 \Delta m_i \right) \omega$$

$$L_{top} = \left(\int r^2 dm \right) \omega \Rightarrow L = I \cdot \omega$$

$$CJ \omega = 10 \text{ devir/sn} \rightarrow 10 \left(\frac{\text{devir}}{\text{s}} \right) \cdot \left(\frac{2\pi \text{ rad}}{1 \text{ devir}} \right) = \frac{20\pi \text{ rad}}{\text{s}}$$

$$m = 7 \text{ kg}$$

$$R = 0.12 \text{ m}$$

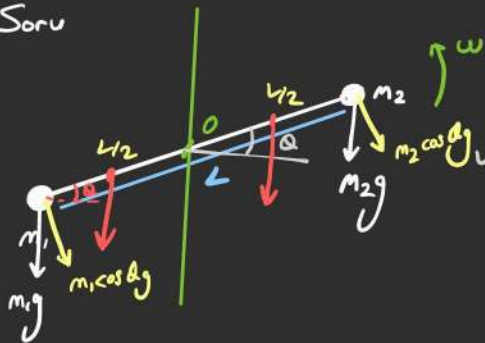
$$L = I \cdot \omega$$

$$\downarrow$$

$$\frac{2}{5} \cdot 7 \cdot (0.12)^2 \cdot 20\pi =$$

$$2.53 \text{ kg} \cdot \frac{\text{m}^2}{\text{s}} =$$

Soru



Açılma momenti.

$$L = I \cdot \omega$$

$$\downarrow$$

$$\left(m_1 \cdot \frac{L^2}{4} + m_2 \cdot \frac{L^2}{4} + \frac{ML^2}{12} \right) \cdot \frac{3(m_1 + m_2)L^2 + ML^2}{12} \cdot \omega = L$$

θ açısı anında sistem dengeli iken bul.

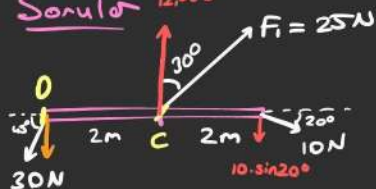
$$\tau = I \cdot \alpha$$

$$|m_1 \cos \theta g - m_2 \cos \theta g| = \frac{3(m_1 + m_2)L^2 + ML^2}{12} \cdot \alpha$$

$$\sum \vec{\tau} = \frac{dL}{dt}$$

$$\sum \vec{\tau} = 0 \text{ ise } \text{açılma momentumu sabit}$$

Soru 10



xy düzleminde hareket eden

1.5kg kütleli cismin hız vektörü

$$(4.2\hat{i} - 3.6\hat{j}) = \vec{v}$$

Orijinden çizilen konum vek = $(1.5\hat{i} + 2.2\hat{j})$

- Açısal Momentum

$$\vec{p} = (6.3\hat{i} - 5.4\hat{j})$$

$$L = I \cdot \omega$$

$$\vec{r} = (1.5\hat{i} + 2.2\hat{j})$$

$$L = \vec{r} \times \vec{p}$$

$$\vec{L} = -20.43\hat{k}$$

$$L = I \cdot \omega = \vec{r} \times \vec{p}$$

$$mV = (m+M) v_{\text{top}}$$

$$\frac{mV}{m+M} = v_{\text{top}}$$

$$(m+M) v_{\text{top}} \cdot l$$

$$r\omega = v$$

$$\omega = \frac{v}{r}$$

Statik Denge ve Esneklik

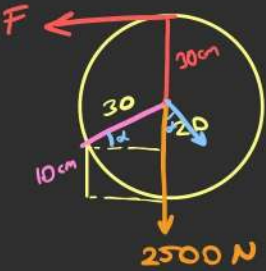
* Tüm kuvvetlerin bileşkesi sıfır ise cisim dengededir.

$$\sum F = 0 \begin{cases} \text{durgun} \\ \text{sabit hız} \end{cases}$$

* Statik = Durgun

Cisim $\begin{cases} \text{noktasal} \rightarrow \sum F = 0 \text{ yeterlidir.} \\ \text{rijit (gerçek)} \rightarrow \sum F = 0 \text{ ve cismin herhangi bir noktaya göre torku sıfır olmalıdır.} \end{cases}$

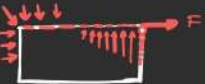
$$\begin{aligned} \underbrace{\sum \vec{F} \cdot \vec{d}}_{\text{öteleme dengesi}} &= \vec{F} \cdot \vec{d} \quad (F \perp d) = \vec{F} \times \vec{d} \\ \underbrace{\sum \vec{F} \cdot \vec{d} \cdot \sin \alpha}_{\text{dönme dengesi}} &= \vec{F} \cdot \vec{d} \cdot \sin \alpha \end{aligned}$$



SOF =

Esneklik

* Gerçekten bütün cisimler bozunmaya uğrar.



* Her bir rijit cisim, dış kuvvetlere karşı belirli bir direnme ve rijitliğini koruma sınırı vardır. Direnme noktasını geçilmemesi için dikkatli davranılmalıdır.

Esneklik Modülü

* Young modülü: Katıların uzunluğundaki bir değişime karşı gösterdiği direnç oranı.

* Kesme sabiti: Katıların, atomik düzeylerde birbirini üzerinde kayması -enine kesme- ortaya çıkan direnç.

* Hacim modülü: Hacim değişikliklerinde gelebilecek direnç ölçütü.

$$F_s = -k \cdot x$$

$$x = A \cdot \cos(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega}$$

$$a_{\max} = A\omega^2 \rightarrow \text{maks. genlik}$$

$$v_{\max} = A\omega$$

$$v(t) = -A\omega \sin(\omega t + \phi)$$

$$a(t) = -A\omega^2 \cos(\omega t + \phi)$$

yay-kitle

$$2\pi \sqrt{\frac{m}{k}} = T = \frac{2\pi}{\omega}$$

anlık kuvvet $\rightarrow$ yollar 30
acisal momentum korunur.

ilerleyen cismin
kaymadan yuvarlan
 $f=0$

Gizise 1

$$m v_0 = m v + m \omega r$$

$$= m \omega d + m \omega r$$

$$r = \frac{L}{2}$$

Acisal

$$L_i = L_s \quad L = r \times p$$

$$m v_0 d = I_h \cdot \omega$$

$$m v_0 = \frac{\left(\frac{1}{3} m L^2 + m d^2\right) \cdot \omega}{d}$$

$$\frac{\left(\frac{1}{3} m L^2 + m d^2\right) \omega}{d} = m \omega (md + mL)$$

$$\frac{1}{3} m L^2 + m d^2 = m d + m L$$

$$\frac{1}{3} L^2 = \frac{1}{2} L$$

$$d = \frac{2L}{3}$$